

A Reconfigurable Embedded Platform for Accurate Ultrasonic TOF Estimation

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Abstract—This paper presents the design and evaluation of a reconfigurable, low-cost embedded platform that integrates an analog front end and digital signal processing for ultrasonic time-of-flight (TOF) estimation. Leveraging the flexibility of the PSoC 5LP architecture, the platform combines configurable analog blocks with real-time digital processing to enable precise TOF measurements using airborne ultrasonic transducers.

Several TOF detection techniques are implemented and evaluated under controlled experimental conditions, including the Akaike Information Criterion (AIC), standard cross-correlation, and a novel envelope derivative correlation method (CEED). The CEED method enhances high-slope regions of the echo signal and suppresses multipath interference, improving robustness in noisy environments.

The results demonstrate that the proposed hardware/software platform achieves submillimeter accuracy in real-time, confirming its potential as a scalable and accessible solution for smart sensing applications such as structural health monitoring, industrial automation, and environmental measurement.

Index Terms—Embedded Systems, PSoC, Analog and digital signal platform, Ultrasound, Narrowband Ultrasonic Sensors, Time-of-Flight (TOF), Cross-Correlation, Envelope Derivative Template.

I. INTRODUCTION

Embedded systems have become a cornerstone in the development of innovative technological solutions. From medical devices to industrial automation systems, these systems enable the efficient integration of hardware and software to perform specific functions autonomously and reliably. Smart sensors, in particular, play a crucial role in providing accurate, real-time data that facilitate automatic decision-making and enhances system performance across various applications. Implementing these sensors within low-cost digital architectures opens new opportunities to expand accessibility and scalability[2]. In this context, platforms like the PSoC 5LP by Infineon, known for their flexibility and affordability, offer a promising solution for designing efficient smart sensors and embedded systems. The digital architecture based on PSoC 5LP allows for the integration of complex functionalities within a single chip, reducing costs and size while maintaining performance [24].

Building on this foundation, this paper presents a practical implementation: a complete embedded platform for ultrasonic TOF measurement using a low-cost, reconfigurable architecture. By showcasing this application, we demonstrate how

a low-cost digital architecture based on the PSoC 5LP can serve as an effective platform for developing smart sensors. Ultrasonic TOF measurement represents a practical and widely adopted method for precise distance measurement and environmental sensing, reflecting the potential of embedded systems to facilitate cost-effective, high-performance sensor solutions. The main contribution of this work is the design and validation of a compact embedded platform that leverages the PSoC 5LP's configurability to implement TOF estimation methods directly in hardware, enabling real-time performance at low cost. A new TOF detection method is also evaluated, but its detailed formulation is left for future work.

This paper is organized as follows. Section II introduces the fundamentals of the TOF detection methods evaluated on the proposed platform. Section III describes the architecture and implementation of the platform and its subsystems. Section IV presents the experimental results and discusses the performance of the implemented methods.

II. TIME-OF-FLIGHT DETECTION METHODS FOR AIRBORNE ULTRASOUND

To illustrate the functionality, versatility, and precision of the proposed platform, this manuscript presents a case study on the estimation of the TOF of airborne ultrasonic waves. Accurate TOF measurement is a fundamental technique used in various applications such as distance measurement, object detection, level measurement in tanks, structural health monitoring, and even medical imaging [8]. The ability to reliably measure the time it takes for ultrasonic waves to travel between sender and receiver enables numerous industries to develop cost-effective and efficient solutions for environmental sensing, automation, robotics, and more. By focusing on an ultrasonic signal, this study also explores the signal conditions that are common in challenging environments, demonstrating the versatility and practicality of low-cost embedded systems for a wide array of applications [17]. These characteristics are essential to ensure precise measurements in practical applications, even under challenging signal conditions.

Two of the methods selected for this study are well known in the literature for their precision and reliability in ultrasonic TOF estimation [5] and serve as benchmark techniques for evaluating the performance of a novel TOF detection method introduced in this work.

Based on these criteria, the following methods were considered:

Akaike Information Criterion (AIC): An established statistical approach adapted for TOF detection, known for its

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robustness in signal analysis. Since it is a statistical method, it does not require a template or prior information about the echo signal [1].

Cross-Correlation: A classic method widely used for its effectiveness in locating delays between signals. This approach relies on prior knowledge of the template or shape of the echo signal [20].

Correlation of the Echo Envelope Derivative (CEED): An innovative algorithm designed to optimize resource use while maintaining accuracy, tailored for embedded implementation.

Both AIC and cross correlation are popular due to their proven precision and reliability, though they involve substantial calculations. The CEED method aims to balance these qualities with computational efficiency for embedded systems.

A. AIC

This approach is based on the Akaike Information Criterion (AIC) picker[1]. The fundamental assumption is that the signal can be divided into locally stationary segments, where the intervals before and after the onset time are treated as two different stationary datasets. For a given order of autoregressive process, the minimum point of the time sequence processed by the AIC algorithm represents the separation point between these two different time series segments, which corresponds to the TOF point[30]. The function is defined as follows:

$$AIC(k) = k \log_{10}(\text{var}(s[1 : k])) + (N - k - 1) \log_{10}(\text{var}(s[k + 1 : N])) \quad (1)$$

where N is the number of data samples in the received signal and $s(1, k)$ and $s(k + 1, N)$ are the signal segments before and after the k th point. The terms $\text{var}(s(1, k))$ and $\text{var}(s(k + 1, N))$ represent the variances of the two signal segments [5].

The AIC function of the echo signal produces a curve with a negative slope. A change in slope occurs at the moment the echo begins, and it is crucial to accurately identify this point of slope change to measure the distance between the receiver and transmitter transducers. To identify this point of slope change, the curve of the AIC function is often derived, resulting in a signal where the point of maximum amplitude corresponds to the onset echo signal.

The principal advantage of the AIC method lies in its template-free approach to TOF estimation [27]. AIC operates solely on the received signal to determine the time of flight. This self-sufficient characteristic makes the method particularly efficient and practical in applications where reference signals are difficult to obtain or may vary significantly between measurements [29]. However, a known drawback of AIC is its sensitivity to noise, especially in low signal-to-noise ratio conditions, which can lead to variability and reduced accuracy in TOF estimation [6].

B. Cross correlation of the echo signal

The cross-correlation method is a fundamental technique widely used to determine time delays between signals[9], particularly in acoustic or ultrasonic time-of-flight (TOF) measurements. The method operates by systematically comparing

a reference signal, $s_T(t)$, with a received signal, $s_R(t)$, at different time shifts[22], [20]. For digital signals of finite length, this comparison is mathematically expressed as:

$$c[n] = \sum_{k=-\infty}^{\infty} s_T[k] \cdot s_R[k - n] \quad (2)$$

where $c[n]$ represents the correlation value at shift n . The time delay between transmission and reception corresponds to the position where $c[n]$ reaches its maximum value, indicating optimal signal alignment[21].

Cross-correlation offers several key advantages for TOF estimation: it provides excellent noise immunity through signal averaging[18], and can achieve subsample time resolution when combined with interpolation techniques[32]. However, this method also presents limitations: it requires prior knowledge of the transmitted signal as a reference template, which may not always be available or may change due to dispersion or attenuation effects[26].

C. Correlation of the echo envelope derivative (CEED)

Traditional cross-correlation exhibits limitations when applied to low-bandwidth signals. A recently developed method introduced by the authors dubbed CEED introduces two enhancements: (1) modifying the transmitted pulse using destructive self interference, and (2) performing cross-correlation on the derivative of the echo envelope. This emphasizes high-slope regions and suppresses multipath ambiguity. Destructive self interference introduces a distinct Gaussian-like echo marker, improving correlation contrast [14].

We present the rationale behind our method and explain its advantages over conventional cross-correlation techniques. As a matched filter, cross-correlation emphasizes signal regions with high amplitude and consistent phase, shaping the output according to these features. When applied to signal envelopes, phase information is neglected, improving robustness but reducing temporal resolution [20]. However, this approach presents a significant limitation when applied to low-bandwidth signals: High-amplitude regions often contain multiple peaks of similar magnitude due to multipath echoes, leading to inaccuracies in signal-onset detection, particularly under low signal-to-noise ratio (SNR) conditions and resulting in distance estimation errors on the order of several millimeters.

The derivative-based approach fundamentally transforms this paradigm. By computing the derivative of the signal envelope, the method reweights the importance of points based on the rate of change in amplitude. In this way, regions of rapid amplitude change become maxima, while flat, high-amplitude regions are de-emphasized. Thus, the method highlights the steepest slope regions of the signal, which are inherently more robust to noise. As mentioned above, the point of maximum slope in the envelope corresponds to a well-defined peak in the derivative signal, providing a more distinct and reliable reference for TOF estimation. When this derivative signal is used in the correlation process, the resulting weighted averaging favors these slope-dominant regions, yielding improved precision and accuracy in TOF detection under noisy conditions; a more

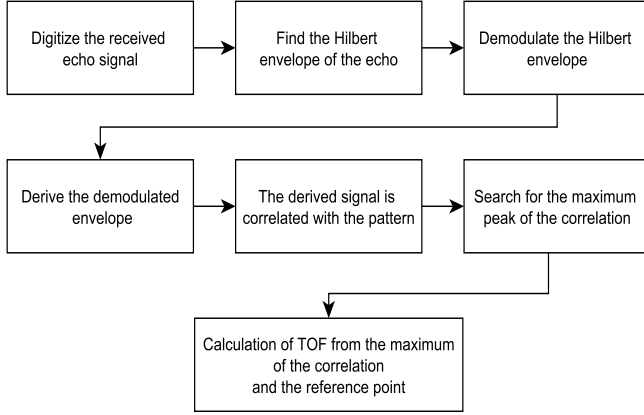


Fig. 1: Signal processing block diagram for TOF estimation based on echo envelope derivative cross-correlation.

detailed formulation and validation will be presented in a separate study.

To further enhance noise immunity and address the inherent bandwidth limitations of the transducer, a destructive interference technique was implemented to shape the echo waveform. This was achieved by generating a secondary excitation signal with a 180-degree phase shift relative to the primary signal, resulting in a more Gaussian-like echo profile as it propagates through the medium. The transformed Gaussian waveform provides two main advantages: increased resistance to environmental noise and improved correlation performance as a result of the more pronounced variations in the envelope's slope. This distinctive temporal marker introduces richer features into the received signal, enabling more reliable and precise TOF measurements. Fig. 2 illustrates the sequence of processing steps used to obtain the correlation template from the received signal. The same processing is applied to the received signal prior to executing the correlation operation.

III. PLATFORM FOR DATA ACQUISITION AND SIGNAL PROCESSING

This section describes the proposed integrated platform for implementing ultrasonic signal detection for TOF estimation, encompassing both hardware and software components. The designed experimental platform serves to evaluate and compare the performance of the different signal detection techniques for time-of-flight estimation previously discussed. In this testbed, airborne ultrasonic transducers are used to experimentally validate each TOF estimation method.

A. Processing architecture

The following criteria were considered when selecting the digital architecture for the experimental tests:

- Low relative cost.
- Scalable and reconfigurable platform architecture.
- Programmable and reconfigurable analog and digital blocks with flexible automatic routing.
- High-performance analog-to-digital conversion.

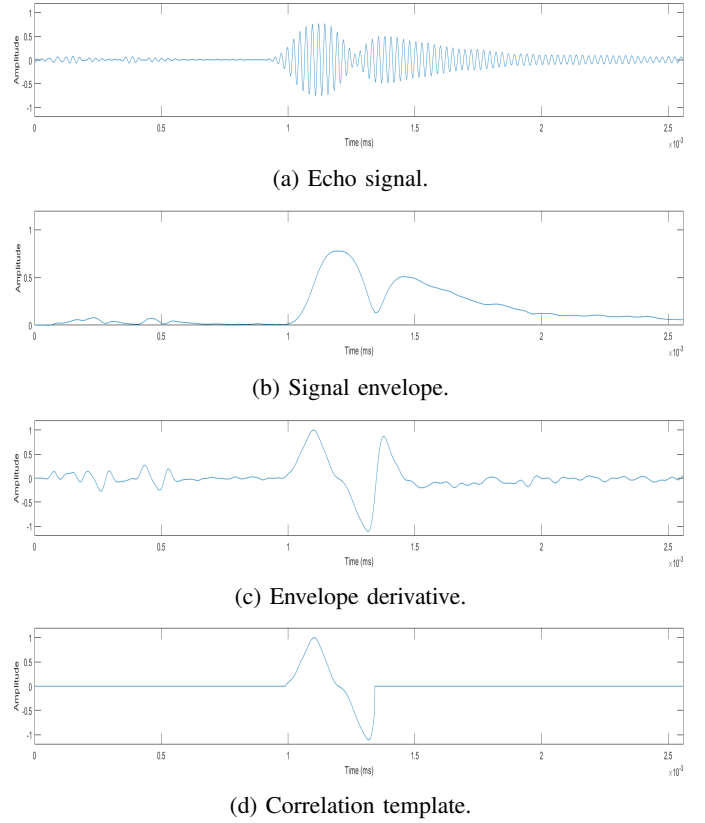


Fig. 2: Processing sequence for the correlation of the echo envelope derivative (CEED) method: (a) echo signal with destructive interference, (b) signal envelope, (c) envelope derivative, and (d) correlation template.

• Fast task execution.

The Cypress PSoC 5LP microcontroller [11] was selected as a suitable solution due to its cost-effectiveness and reconfigurable architecture. This device integrates programmable analog and digital blocks with a flexible routing system that enables peripheral modifications without physical hardware changes[10]. The embedded ARM Cortex-M3 processor[4] provides sufficient performance and efficiency for analog-to-digital conversion tasks.

The versatility and adaptability of the PSoC 5LP make it particularly suitable for prototypes that require scalability, processing speed, and frequent design iterations. The selected processing architecture enables on-chip integration of the excitation, conditioning, acquisition, and signal processing circuits.

B. Ultrasonic transducer and excitation circuit

For the experimental tests, Murata MA40S4R airborne ultrasonic transducers [28] were selected, which feature a nominal resonance frequency of 40 kHz. These piezoelectric transducers offer key characteristics suitable for TOF measurement applications, including a directional beam pattern with approximately 60° beam width at -6 dB, sensitivity specifications of -65 dB (receiving), and sound pressure level of 100 dB at 10 cm (transmitting). Their compact form

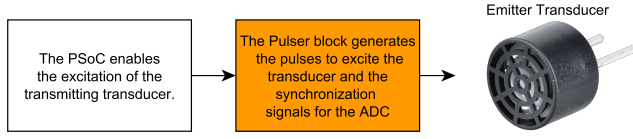


Fig. 3: Block diagram illustrating the transducer excitation procedure. Hardware-implemented operations are shown in orange, while software-controlled processes are shown in white.

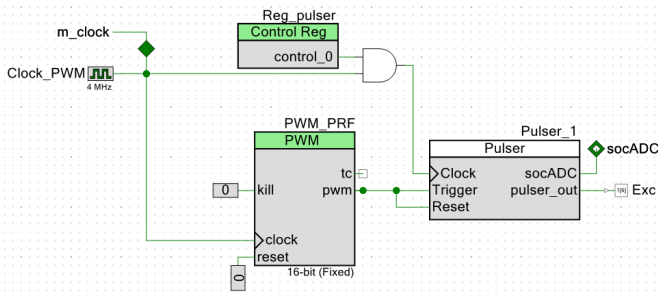


Fig. 4: Implemented transducer excitation circuit.

factor of 16 mm diameter facilitates easy integration into the experimental setup while maintaining measurement accuracy.

The excitation circuit was fully implemented using internal PSoC components, as shown in Fig. 4. This circuit includes a custom component written in Verilog, known as the “Pulser.” The Pulser generates square pulse trains to excite the transmitting transducer. It is controlled by a control register, allowing the microprocessor software to manage it, and by a PWM module used as a trigger to synchronize the pulse emission with the start of signal digitization. This hardware generates the excitation pulse to produce the self-interference in the emission process; see Fig. 2a.

C. Signal acquisition and processing circuit

Fig. 5 illustrates the CEED procedure used to estimate the TOF. It is important to note that a significant portion of the digital signal processing is performed in hardware, which enhances the process efficiency.

The airborne ultrasonic transducers used in this project are tuned to a resonance frequency of 40 kHz. The signal conditioning and digitization circuit, shown in Fig. 5, was fully implemented using the internal components of the PSoC architecture. The analog conditioning stage consists of a differential charge amplifier implemented with two transimpedance amplifiers (TIAs). The conditioned signal is then digitized using a differential-mode successive approximation register (SAR) analog-to-digital converter (ADC), operating at a sampling frequency of 400 kHz. To correctly digitize the echo signal, it is biased at 2.5V DC using an internal operational amplifier, as the PSoC’s ADC component is configured to digitize only positive voltages. To ensure precise synchronization, the ADC

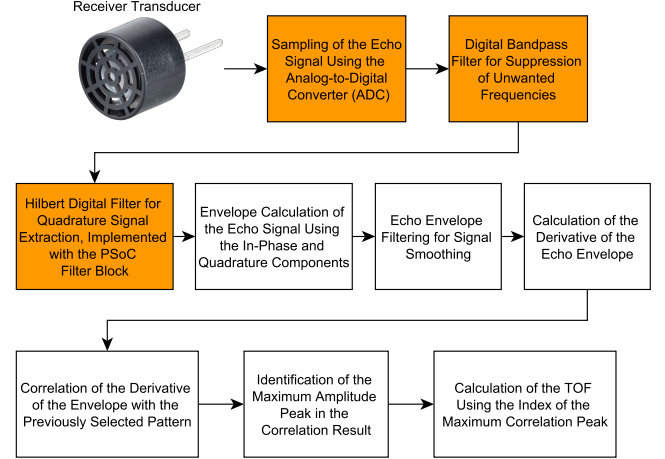


Fig. 5: Block diagram illustrating the CEED method. Hardware operations are shown in orange; software processes in white.

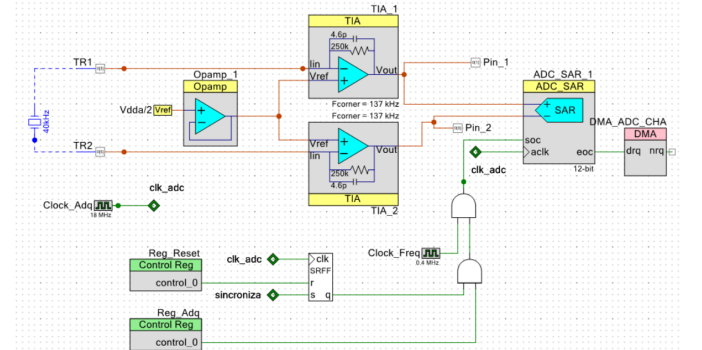


Fig. 6: Signal conditioning and digitization circuit implemented within the PSoC platform.

is controlled by two AND logic gates, a flip-flop, and two control registers.

In the CEED method, once the echo signal is digitized, a Direct Memory Access (DMA) module transfers the data to the first processing stage, illustrated in Fig. 7. The design strategically offloads computationally intensive signal processing tasks—such as bandpass filtering and envelope detection—to dedicated hardware blocks within the PSoC, significantly enhancing efficiency. The bandpass filter, centered at 36 kHz, suppresses out-of-band noise and is followed by a Hilbert filter to generate the quadrature component of the signal, see Fig. 7. The envelope of the signal is computed by taking the square root of the sum of the squares of its in-phase and quadrature components by the microcontroller. Following, the envelope is processed for the time-of-flight (TOF) estimation.

The remaining processing is performed by the PSoC’s ARM Cortex-M3 microprocessor. For AIC and cross-correlation methods, the hardware involved is limited to the signal acquisition blocks; see Fig. 6.

In addition, an AM2315[3] sensor is connected to the PSoC

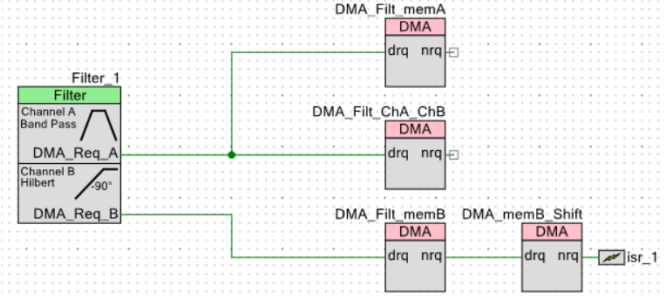


Fig. 7: Digital filters implemented in hardware after signal digitization for in-phase and quadrature component extraction.

to measure the temperature of the medium, allowing the calculation of the propagation speed of the mechanical wave.

IV. EXPERIMENTAL EVALUATION AND RESULTS

To evaluate TOF estimation methods, a test bench was assembled using two 40 kHz ultrasonic transducers (one transmitter and one receiver) positioned face to face. The transducers were mounted on a calibrated optical rails (Newport PRL-36) to ensure precise alignment and accurate distance control.

The AIC and echo cross-correlation methods are considered as gold standards for evaluating the performance of the CEED method. To assess the precision and accuracy of the proposed techniques, two experiments were conducted. In the first experiment, a data set of 100 echo measurements was collected with the transmitter and receiver positioned at a fixed distance of 300 mm. The resulting distance estimations are presented in Fig. 8. It can be observed that the echo correlation method yields relatively consistent results but produces errors of up to 8 mm. This maximum error corresponds to a single outlier, likely influenced by the wavelength of the echo signal. The AIC method shows high sensitivity to noise, fails to maintain measurements within the expected range, and exhibits errors exceeding 7 mm. In contrast, the correlation of the echo envelope derivative (CEED) method demonstrates superior robustness, with a maximum error below 3 mm.

In the second experiment, the transducers were initially positioned 400 mm apart. Ten distance measurements were then performed at 10 mm intervals, incrementally increasing the separation up to 600 mm. The reference point (0 mm offset) was defined at a separation of 500 mm between the transducers. Based on the collected data, the mean error and standard deviation were calculated for each distance interval. The results are presented in Fig. 9.

The AIC method exhibits a moderate standard deviation but consistently yields a higher average error compared to the other techniques. The echo correlation method produces an average error close to zero; however, its standard deviation increases with distance, reaching up to 4 mm. In contrast, the correlation of the echo envelope derivative (CEED) method maintains both a very low average error, below 1.5 mm and a standard deviation also under 1.5 mm throughout the entire range. The corresponding average values are summarized in Table I, and the standard deviations are presented in Table II.

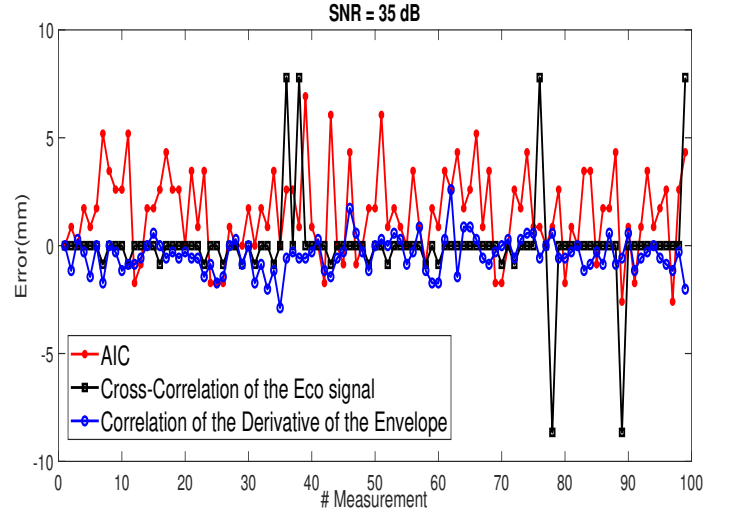


Fig. 8: Comparison of TOF estimation errors using AIC, echo correlation, and CEED methods at an SNR of 35dB. Results are based on 100 measurements with the transducers fixed at a distance of 300 mm.

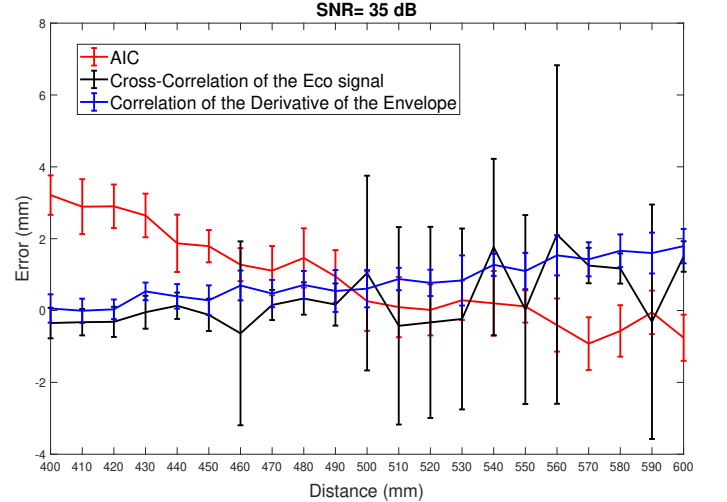


Fig. 9: Distance measurement results for different TOF estimation methods across varying transducer separations (400 mm to 600 mm) at an SNR of 35 dB.

V. CONCLUSION

This work demonstrates the design of a low-cost, reconfigurable embedded platform capable of accurate ultrasonic time-of-flight (TOF) estimation. Leveraging the flexibility of the PSoC 5LP, the system integrates analog signal conditioning, digitization, and real-time processing within a compact and efficient embedded device. The proposed CEED method outperformed traditional techniques such as AIC and echo correlation under noisy conditions, achieving sub-millimeter precision with low standard deviation. The modular design of the platform allows future adaptation for other sensing modalities, leveraging the same processing architecture.

Reference (mm)	AIC (mm)	Echo Correlation (mm)	Derivative Correlation (mm)
-100	-96.787	-99.651	-99.536
-80	-77.099	-79.686	-79.57
-60	-58.129	-59.867	-59.956
-40	-38.723	-39.365	-39.767
-20	-18.536	-19.665	-19.666
0	0.260	1.043	0.173
20	19.983	19.669	19.636
40	39.798	38.234	38.988
60	59.595	59.536	58.869
80	79.431	78.830	78.830
100	99.242	98.497	98.758

TABLE I: Average distance measurement errors at an SNR of 35 dB. Corresponding results are illustrated in Fig. 9

Reference (mm)	AIC (mm)	Echo Correlation (mm)	Derivative Correlation (mm)
-100	0.548	0.425	0.309
-80	0.607	0.419	0.362
-60	0.798	0.366	0.238
-40	0.458	2.56	0.419
-20	0.824	0.448	0.381
0	0.260	2.711	0.436
20	0.709	2.661	0.469
40	0.898	2.458	0.311
60	0.738	4.694	0.381
80	0.715	0.419	0.372
100	0.642	0.423	0.506

TABLE II: Standard deviation of distance measurements at an SNR of 35 dB, corresponding to the results shown in Fig. 9.

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